DESIGNING AN INTEGRATED MULTI-DIMENSIONAL ASSESSMENT FRAMEWORK FOR AI-SUPPORTED ACADEMIC WRITING

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ABSTRACT

The widespread adoption of generative AI is transforming academic writing in higher education, rendering traditional, product-focused assessment models obsolete. These methods fail to capture the iterative and tool-mediated nature of modern writing processes, creating an urgent need for new evaluation approaches. This paper addresses this gap by proposing an integrated, multi-dimensional assessment framework. Grounded in genre pedagogy, self-regulated learning, and writing analytics, our model conceptualizes assessment as a holistic process. It evaluates foundational skills for AI use (Computational Thinking and Genre-Knowledge), the quality of text revision, final writing performance, and long-term development. By aligning these dimensions with formative, summative, and diagnostic purposes, the framework fosters transparency, metacognitive engagement, and responsible AI use. We are preparing a design-based research project to pilot and iteratively refine key elements of the framework within a mastery learning programme for 1,800 first-year students. The goal is to explore how the model can be implemented in practice and refined through iterative cycles of design, enactment, and evaluation.

KEYWORDS

Generative AI, Academic Writing, Writing Analytics, Evidence-Centered Design (ECD), Computational Thinking (CT)

1. INTRODUCTION

The emergence of generative artificial intelligence (GenAI), particularly large language models (LLMs) such as ChatGPT, Claude, and Gemini, has rapidly transformed the landscape of academic writing in higher education (Malik et al., 2023; Spirgi et al., 2024). Students now widely use such tools to support a range of writing tasks from idea generation and text structuring to paraphrasing and full-text drafting (Malik et al., 2023; Spirgi et al., 2024). These capabilities offer considerable potential to support learning and enhance writing processes. However, they also bring forth critical questions regarding authorship, cognitive engagement, academic integrity, and the validity of traditional assessments, underscoring the "profound paradox" (Gulumbe et al., 2024, p. 1775) of AI in scholarly pursuits (Malik et al. 2023).

Academic writing has long been recognised as a complex, socially situated, and cognitively demanding activity that evolves over time and across contexts (Hyland, 2003). In light of GenAI's capabilities to automate or augment many aspects of writing, there is an urgent need to reconceptualise what it means to assess writing in educationally meaningful and ethically responsible ways (Gulumbe et al., 2024; Suglo et al., 2024). Existing assessment models, often focused on the final product or switching to more oral assessments, are increasingly insufficient for capturing the full spectrum of learner activity and development in AI-supported writing environments (Hau, 2025; Khlaif et al., 2025; Lodge et al., 2023).

This paper addresses that gap by proposing a conceptual, multi-dimensional framework for the assessment of academic writing in the age of GenAI. The framework is designed to account for both the product (e.g., textual quality, genre conformity) and the process (e.g., revision behaviour, the application of Computational Thinking (CT) and Genre-Knowledge during AI interactions) of writing, as well as the human-AI interaction that increasingly shapes students' writing trajectories (Yang et al., 2024).

The central aim of this study is to develop a theoretically robust, pedagogically meaningful, and practically applicable framework for the assessment of academic writing in AI-supported learning environments. The proposed model is designed to reflect the evolving practices, literacies, and challenges associated with GenAI in higher education. To guide its development, the paper addresses the following research questions:

1) What key dimensions should be included in a pedagogically grounded framework for assessing AI-supported academic writing?
2) How can each dimension be aligned with appropriate forms of evidence and assessment purposes (formative, summative, diagnostic)?
3) In what ways can GenAI meaningfully contribute to both writing support and the assessment process?

The paper proceeds as follows: Chapter 2 reviews theory, Chapter 3 outlines methods, Chapter 4 presents the multi-dimensional model, and Chapter 5 illustrates its university implementation through digital modules and AI tools. The last two chapters evaluate strengths, limits, implications, and future work, aiming for a holistic, transparent, educationally aligned assessment of academic writing in the AI era.

2. THEORETICAL BACKGROUND

2.1 Generative AI and Genre-Based Academic Writing

In academic writing courses, GenAI tools like ChatGPT now support students in various tasks, from brainstorming to drafting full texts (Spirgi & Seufert, 2024). Tools based on large language models (LLMs), such as ChatGPT, Claude, or Gemini, now support students in various writing tasks such as brainstorming, outlining, paraphrasing, translating, and even drafting full academic texts (Spirgi et al., 2024). These technologies offer new opportunities for writing support, especially for learners with limited experience in academic discourse or language proficiency (Kasneci et al., 2023). Their integration raises critical challenges regarding overreliance, originality, and academic integrity (Dehouche, 2021; Deng et al., 2025; Lund et al., 2025; Zhai et al., 2024).

Rather than viewing the use of GenAI through the narrow lens of acceptance or rejection, recent pedagogical discourse emphasises the need for a critical, reflective and intentional approach to integration (Nguyen, 2025). According to Eager & Brunton (2023) and Gattupalli et al. (2023), prompting is an academic skill requiring explicit teaching and assessment. Creating effective prompts demands articulating goals, understanding context, and engaging iteratively with AI outputs as a rhetorical act.

Effective academic writing with GenAI requires more than just good prompting; it necessitates a broader CT approach. Unlike prompt literacy (Gattupalli et al., 2023), which centres on the art of formulating individual queries, CT provides a more comprehensive and strategic framework that orchestrates the entire AI-assisted writing process. CT, drawing on computer science fundamentals (Wing, 2006), involves problem-solving and is a cognitive process encompassing abstraction, decomposition, algorithmic thinking, evaluation, and generalisation (Selby & Woollard, 2013). Crucially, applying CT to AI-assisted academic writing demands specific contextual understanding, where genre-knowledge is vital. Our framework views CT not just as prompt formulation, but as metacognitive planning of the AI-writing process, decomposing tasks, critically evaluating AI outputs against genre conventions, and iteratively refining prompts based on error analysis.

This connection reinforces the relevance of genre-based pedagogy, which conceptualises academic writing as socially situated and discipline-specific (Hyland, 2003). Genres function as scaffolds that structure both textual production and interpretation, guiding students in navigating complex communicative demands (Hyland, 2003; Spirgi et al., 2024). Genre-based instruction integrates product- and process-oriented perspectives (Badger & White, 2000), enabling students to understand not only how to write, but also why and for whom. This perspective is particularly important in AI-supported writing, where intentionality and discourse conventions shape effective tool use.

Applying CT principles such as decomposition of tasks according to genre conventions or pattern recognition within disciplinary texts allows students to strategically engage with AI. It moves them beyond simple text generation towards a more informed and critical interaction with these tools, aligning AI output with the nuanced expectations of academic discourse (Celik, 2023). In sum, the convergence of GenAI and genre-based pedagogy opens new pedagogical horizons. It invites educators to teach writing not only as text production, but also as strategic tool use informed by disciplinary discourse norms. Supporting students in developing this genre-awareness offers a promising pathway to enhance both their writing and their engagement with AI in academically meaningful ways.

2.2 Rethinking Assessment in AI-Supported Writing

Traditional writing assessment, with its focus on the summative evaluation of final texts, falls short in AI-supported contexts as it fails to capture the iterative, collaborative, and tool-mediated nature of the writing process. It fails to consider how students interact with, reflect on, and revise using AI systems, which are key indicators of learning and competence (Lodge et al., 2023). Current scholarship advocates for multi-dimensional assessment models that integrate product and process views, capture revision practices, and assess human-AI interaction (Cheng et al., 2024; Wang et al., 2025). In this context, assessment must also consider CT skills; revision behaviour across drafts; interaction quality with AI tools; ethical attribution of AI-generated content; reflective engagement with writing processes.

Such models emphasise formative assessment and self-regulated learning (SLR), offering opportunities for feedback, transparency, and developmental insight. To ensure pedagogical relevance in AI-enabled writing contexts, a comprehensive framework must draw on evidence from both the final text and the underlying writing process.

2.3 Learning Analytics and Writing Analytics

Learning Analytics (LA) and Writing Analytics (WA) are essential for assessing AI-supported writing. LA tracks learner behaviour over time, while WA analyses texts and revision processes to provide targeted feedback (Buckingham Shum et al., 2016; Siemens, 2013). WA bridges product and process assessment through computational analysis of coherence, genre conformity, and revision quality (Shibani et al., 2019).

Automated Essay Scoring (AES) systems have evolved from rule-based methods (Attali & Burstein, 2004; Shibani et al., 2019) to machine learning (Chen & He, 2013; McCarthy & Jarvis, 2010; Yannakoudakis & Briscoe, 2012) and modern transformer-based (Devlin et al., 2019) approaches, with LLMs now offering sophisticated text understanding capabilities (Li & Ng, 2024). These tools enable formative assessment at scale and can trace human-AI interactions, providing insights into student engagement and competence development.

However, implementing these analytics raises ethical concerns about authorship, data privacy, and feedback interpretability. Effective deployment requires pedagogically grounded frameworks that ensure fairness, transparency, and genuine learning support. The integration of WA with GenAI necessitates viewing writing as a socio-cognitive, tool-mediated process requiring ethical guidelines and reflective practices.

3. METHODOLOGICAL APPROACH

This study follows a conceptual research design, presenting the foundational framework that constitutes the initial phase of a broader design-based research project for assessing AI-supported academic writing through iterative literature analysis, drawing on SLR (Zimmerman, 2002), genre theory, and WA. The framework applies Evidence-Centered Design (ECD) principles (Drachsler, 2023; Mislevy et al., 2003) to ensure methodological rigor. ECD structures each dimension through three components: target construct (what is assessed), evidence model (valid performance indicators) and task model (assessment activities). This approach ensures transparency and pedagogical alignment in AI-mediated writing assessment.

4. CONCEPTUAL OVERVIEW

This chapter presents a conceptual framework (cf. Figure 1) for assessing academic writing in AI-supported learning environments that fundamentally reconceptualizes writing assessment through the lens of SRL. Rather than treating AI as merely a tool to be monitored, our framework positions it as an integral component of the contemporary writing process, requiring new forms of assessment that capture both human competence and human-AI interaction quality.

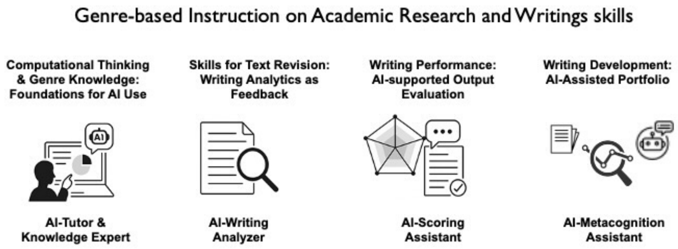

Figure 1. Integrated, multi-dimensional Assessment Framework for Academic Writing

The framework translates Zimmerman's (2002) SRL model into four linked dimensions that map the full self-regulatory cycle. CT and Genre-Knowledge anchor the forethought phase, guiding strategic planning for AI-mediated writing. Skills for Text Revision captures the recursive performance-reflection loop within a single task. Writing Performance becomes the artifact for holistic self-reflection, while Writing Development tracks longitudinal growth from scaffolded to autonomous strategy use. Unlike traditional models that isolate products or processes, this approach treats assessment as a learning catalyst, documenting and nurturing autonomous, critically engaged writers in the AI era. Its success presupposes a dual-level ecosystem: individual educators who can critically integrate evolving AI tools into pedagogy, and institutions that supply infrastructure and strategic support (Markauskaite et al., 2022).

4.1 Computational Thinking and Genre-Knowledge as Foundations for AI Use

Combining CT and deep Genre-Knowledge is fundamental for strategically planning, executing, evaluating, and refining GenAI use in academic writing as shown in Section 0. GenAI acts as a cognitive tool, its deployment shaped by the learner's CT and genre-knowledge, and can also be used to build declarative genre knowledge. Primarily, formative assessment aims to foster these abilities, as CT provides the procedural framework for GenAI interaction, while genre-knowledge offers contextual understanding for academically sound outcomes. These intertwined competencies are vital for discerning, ethical, and effective AI application, influencing Text Revision Skills and overall Writing Performance, forming the basis for self-regulated engagement with AI.

The assessment framework focuses on the *construct* of a student's proficiency in applying CT principles like decomposition, algorithmic interaction, pattern recognition, abstraction, and systematic evaluation, while also utilising genre knowledge for strategic planning, guidance, and critical revision in AI-assisted writing. The *evidence model* for these integrated skills becomes apparent when students document their method of breaking down tasks for AI, articulate their iterative adjustments based on AI feedback and genre-specific needs, or validate changes to AI-generated text by referencing logical structure and genre conventions, including their reasoning for AI tool selections and prompts that specify genre characteristics. The *task model* employs activities such as AI-supported writing assignments specific to a genre coupled with reflective reports, the critique and revision of AI texts against established genre criteria while explaining the CT principles applied, "AI process walkthroughs" where students describe their step-by-step, genre informed CT approach, and comparative genre analyses using AI to demonstrate their ability to identify underlying patterns and communicative functions.

4.2 Skills for Text Revision

The primary focus is on a student's capability to pinpoint areas for improvement in their writing, effectively incorporate feedback, and implement substantial revisions that enhance the overall quality of the text. Here, WA tools support students by providing targeted feedback on their drafts, empowering them to engage in reflective revision and develop self-regulation skills. The assessment function is primarily formative and diagnostic. It aims to identify strengths and weaknesses to guide further learning rather than just assigning a grade.

The core *construct* being assessed is the student's ability to recognise deficiencies in a piece of writing. This includes their capacity to react constructively to feedback and execute meaningful changes that improve the text's structure, clarity, and the strength of its arguments. *Evidence* of these skills can be gathered through various means. This includes analysing and comparing different drafts of a text, examining annotations made during the revision process, observing how students utilise AI-generated suggestions, and reviewing logs that detail their revision behaviours (such as distinguishing between superficial edits and more profound structural changes). The *task model* for assessing these skills typically involves revision assignments that track different versions of the text. It also includes structured feedback mechanisms, whether from peers or AI, and requires students to resubmit their work with annotations that explain the rationale behind their revision choices.

This dimension targets revision competence, focusing on how students revise rather than just outcomes. It assesses the skills and strategies learners apply during revision, including interpreting feedback, prioritising goals, and implementing improvements. WA tools can classify and visualise revisions, visualise changes between drafts, differentiate surface from structural revisions, and provide guidance, supporting feedback and instructional design. Assessments may involve annotated revisions, revision plans, or comparative draft analyses. The emphasis is on cultivating self-regulated revision competence within human-AI collaboration.

4.3 Writing Performance

The evaluation of the final written product focuses on its quality, adherence to genre conventions, and academic standards. Leveraging WA tools as detailed in Section 0, GenAI supports summative assessment by facilitating pre-submission feedback through automated scoring, text analysis, and genre-based diagnostics. The primary assessment function is summative, determining the final evaluation of the written work. However, it can optionally integrate pre-feedback mechanisms for formative support, aiding the writer's self-reflection and development process.

The core *construct* being assessed is the student's ability to produce a high-quality academic text. This text must align with established genre conventions, adhere to disciplinary norms, and meet rhetorical standards. *Evidence* for this assessment is gathered from several sources: the final submitted texts themselves, rubric-based scoring performed by human evaluators, and AI-supported text analysis. This AI analysis can provide insights into aspects like the text's structure, coherence, lexical variety, and overall genre compliance. The *task model* involves genre-specific writing assignments, such as essays or literature reviews. These tasks are evaluated using detailed rubrics. Optionally, students can benefit from pre-submission feedback loops supported by AI, allowing for iterative improvement before final submission.

This dimension aligns with traditional outcome-based evaluation but is adapted to the new affordances of AI. Final texts are assessed using rubrics and genre-specific criteria, potentially supported by AI-driven text analysis tools. AI tools can provide pre-assessment diagnostics, while human evaluators apply calibrated rubrics. This dual process ensures both consistency and context-aware judgment in assessing final products. Metrics such as coherence, clarity, lexical richness, argument structure, and genre conformity guide the evaluation. Where permitted, AI tools can offer formative feedback before submission. The goal is to maintain academic rigor while enhancing transparency and student agency.

4.4 Writing Development

This section describes the development of writing competence as a continuous process, where AI serves as a supportive tool for iterative drafting and feedback. Assessment should focus on improvement over time, utilising portfolios and version tracking to map the growth of self-regulation.

The assessment function is primarily formative, aimed at guiding ongoing learning and improvement. However, summative elements can be incorporated, particularly in portfolio-based contexts where a collection of work is evaluated. The core *construct* being assessed is the growth in self-regulated academic writing competence over time. This includes an increasing awareness of different genres, the ability to produce coherent texts, and the command of rhetorical strategies. *Evidence* for this development is gathered from version histories of written pieces, reflective annotations made by students, their metacognitive statements about their learning process, and logs of their interactions with AI tools. The *task model* for fostering and assessing this development often involves process portfolios, assignments that require multiple drafts, reflective journals, and other longitudinal writing tasks designed to track progress over an extended period.

Our four-dimensional framework enables a pedagogically sound evaluation of AI-supported academic writing processes, shifting the focus from mere outcomes to the process, interaction, and competence development, thus supporting transparent, formative, and ethical assessment that fosters SLR.

5. IMPLEMENTATION SCENARIO

The implementation of the assessment framework has begun with the first dimension already operational, while the remaining components are in various stages of development and planning at our university as part of a large-scale mastery learning programme for approximately 1,800 first-year students in economics, law, and social sciences. Full implementation of all four dimensions is targeted for completion by September 2026. The programme aims to develop foundational competencies in academic research and writing. The first three of the four framework dimensions are operationalised in a staged and integrative learning design of mastery learning.

1. Computational Thinking and Genre-Knowledge as Foundations for AI Use

In "Brian", a self-paced GenAI chatbot module, students master responsible academic use of GenAI. Interactive vignettes hone critical thinking by guiding them to deconstruct problems and craft queries, while concise genre instruction supports the practice. Mastery is certified by the 80% "AI Literacy License" assessment.

2. Text Revision Skills

In the next phase, students engage in targeted writing exercises focused on revision strategies and the improvement of text quality (diagnostic assessment). Therefore, we design a WA tool, which provides formative feedback on coherence, structure, and language use. A pilot study is planned to test the integration of the WA tool in these exercises. The pilot will explore how students engage with AI-generated feedback and how it can support the development of revision competence in authentic writing tasks.

3. Writing Performance

The student-facing system does not yet include final writing performance evaluation. Currently, an AI-supported pre-correction tool is being developed and tested to provide automated formative feedback. This tool is being piloted with a small group of five lecturers from the broader teaching team (33 instructors total) in late 2025. Once validated, it will roll out to students for rubric-based feedback before final submission, upholding standards while ensuring transparency, fairness, and agency. Instructors gain cohort-level insights to tailor teaching and target support, reinforcing their role as learning guides.

6. DISCUSSION

6.1 Strengths of the Framework

A central strength of the framework lies in its comprehensive and integrative design. It expands beyond traditional product-based assessment by incorporating process-oriented and formative components, thereby aligning with current conceptions of writing as a socio-cognitive, iterative, and tool-mediated practice (Rad et al., 2024). Through its differentiated structure addressing diagnostic, formative, and summative functions, the framework supports diverse pedagogical goals and assessment contexts (Eager & Brunton, 2023; Khlaif et al., 2025; Lodge et al., 2023).

It is both theoretically grounded and technologically responsive, drawing on established insights from genre pedagogy, SLR, and WA, while also integrating the affordances and challenges introduced by GenAI. This dual anchoring ensures conceptual rigor and practical relevance across instructional settings.

The inclusion of CT as a distinct assessment dimension introduces an important innovation. In AI-supported writing environments, academic literacy increasingly encompasses the ability to engage strategically, ethically, and reflectively with generative tools. The framework acknowledges this shift by assessing not only writing outcomes but also students' interaction with AI as part of their cognitive and rhetorical process. By operationalising writing competence through traceable indicators such as revision histories, prompting behaviour, or interaction logs, the framework provides actionable data for learners, instructors, and researchers. Writing performance profiles, generated through a combination of system-based analytics (e.g., linguistic complexity) and learner actions (e.g., prompt refinement, revision depth), offer a nuanced basis for feedback and instructional support.

6.2 Limitations and Challenges

While the proposed framework offers a conceptually coherent and pedagogically grounded approach, its limitations must be clearly acknowledged. The framework is entirely theoretical and has not yet undergone empirical validation. As a result, its practical feasibility, pedagogical impact, and effectiveness in real-world instructional settings remain uncertain. Further research is essential to evaluate its applicability and implications across diverse educational contexts.

A significant challenge lies in the implementation, which depends heavily on institutional support and pedagogical readiness. The model assumes a high level of AI competence among educators, yet it does not detail specific support structures or professional development protocols. For a successful integration, educators require robust training in interpreting WA, scaffolding AI-assisted writing tasks, and fostering responsible AI use among students. Without sufficient resources, the integration risks remaining superficial and ineffective, as teachers must transition from mere instructors to learning guides who can coach students in the critical use of AI tools (Tan et al., 2025).

Furthermore, the framework's reliance on data-driven analytics raises significant methodological and ethical concerns (Azra & Zeeshan, 2025). There is a risk of reductionism, where key dimensions of writing such as creativity, rhetorical flexibility, and disciplinary variation are not fully captured by automated metrics. This leads to ethical questions surrounding the explainability of AI-generated feedback, academic integrity (Cotton et al., 2024; Gulumbe et al., 2024; Lund et al., 2025) and data privacy (Huang, 2023).

To address the latter, it is crucial to implement specific safeguards, including data anonymization, data minimisation, and secure, encrypted storage. While the framework mitigates these risks by combining AI-based analysis with human feedback, its overall success critically depends on addressing the challenge of explainability. A core requirement is therefore the development of comprehensible guidelines to ensure that AI-generated feedback is not only interpretable but also fair, preventing the reinforcement of biases or reductionist assessments.

Finally, assessing the nuances of the dialogic interaction between student and AI remains methodologically challenging. It is difficult to distinguish between strategic engagement informed by CT and Genre-Knowledge versus simple trial-and-error without deeper insight into students' intentions. To avoid mere surveillance, any analysis of interaction logs must focus on qualitative patterns that indicate strategic action (e.g., iterative refinement of prompts) and must be conducted with full transparency and student consent.

7. CONCLUSION AND OUTLOOK

This paper presents a conceptual framework for assessing academic writing in the context of GenAI. By defining four dimensions CT and Genre-Knowledge as Foundations for AI Use, Skills for Text Revision, Writing Performance, and Writing Development the model addresses a holistic shift from product-based to process-oriented writing practices in higher education. The framework integrates insights from writing pedagogy, genre theory, SRL, CT and WA, and supports both formative and summative assessment. It highlights the importance of metacognitive reflection, ethical AI use, and emerging literacies in AI-mediated writing.

Future directions must focus on the empirical validation and practical implementation of the framework. Addressing the limitations outlined in the discussion is a critical priority. This involves operationalising the framework for educators through a robust professional development program and clear implementation guidelines, ensuring that the model can be effectively and ethically adopted. The goal is not only to assess current writing practices but to guide their evolution toward more transparent, reflective, and educationally aligned approaches in the age of AI.

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